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- **Problem Statement ID –** SIH26185
- **Problem Statement Title-** Helmet mounted conformal antenna for tactical communications in urban CQB environments.
- **Theme -** Robotics and Drones
- **PS Category -** Hardware
- **Team ID** - 141223
- **Team Name -** Spectron Sentinels



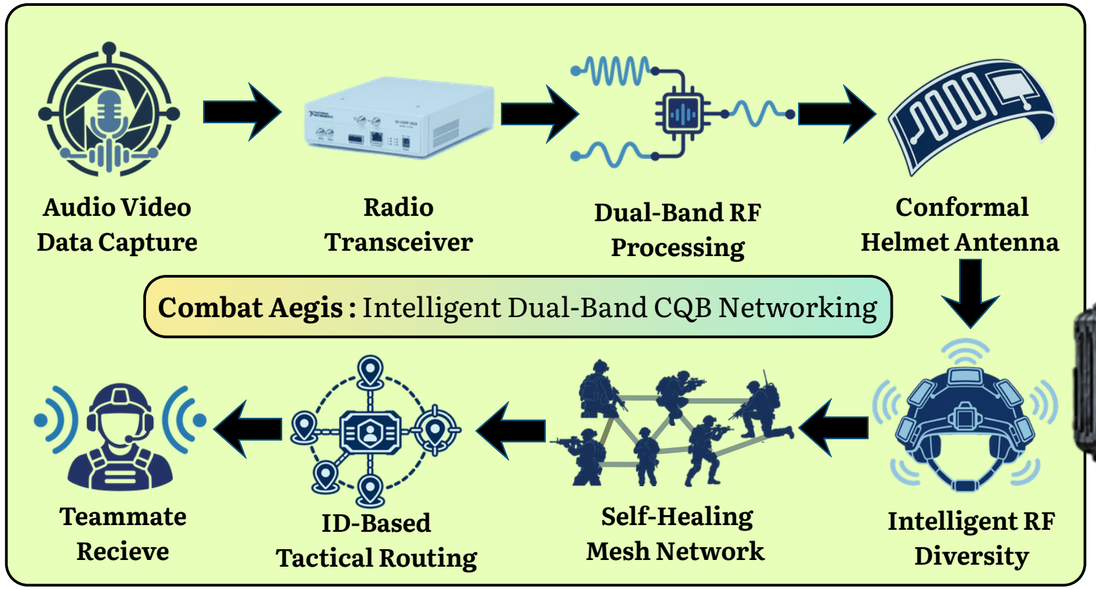
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PROPOSED SOLUTION

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OUR PROPOSED SYSTEM

- **403–470 MHz UHF + 1.55–1.65 GHz** L-band single-sheet conformal antenna for tactical **voice & control** and **high-rate data** links.
- **Four spatially** distributed radiators on a **280 × 40 mm** flexible RF sheet, with ≈240 mm front–rear separation for improved **azimuth** coverage.
- **Single** 50 Ω feed using a **passive diplexer**, 50/50 UHF **Wilkinson** divider and **80/20 L-band** power split-eliminating RF switching.
- Meandered **UHF monopoles** + **AMC**-backed **L-band PIFAs**, optimized for **≤5 mm** profile, **≤25 g mass**, **S11 ≤–10 dB** and helmet & body loading.





INNOVATION & UNIQUENESS

1. Single-Feed Dual-Band Conformal RF Sheet
403–470 MHz UHF + 1.55–1.65 GHz L-band integrated on one flexible **280 × 40 mm** helmet-conformal RF sheet.

2. Spatially Distributed Four-Radiator Architecture
Front/rear UHF + L-band **radiators** with **≈240 mm** separation for **spatially** distributed helmet radiation.

3. Coverage-Optimized Coherent Radiation
S11, **gain**, null-depth & **azimuth** coverage under **CQB multipath** and body-shadowing.

4. Miniaturized Low-Profile Conformal Radiators
Meandered UHF **monopoles** + **AMC-backed** L-band PIFAs targeting **≤5 mm** profile and **≤25 g** total antenna assembly.

5. Passive Frequency-Selective RF Distribution
Diplexer + **50/50** UHF **Wilkinson** + **80/20 L-band** splitter for **single-coax**, **switch-free** RF distribution.

Antenna Design Specifications :

Parameter	UHF	L-Band
Frequency Range	403–470 MHz	1.55–1.65 GHz
Nominal Center	436.5 MHz	1.60 GHz
Bandwidth	≥ 67 MHz	≥ 100 MHz
S11(reflection loss)	≤ −10 dB	≤ -10dB
VSWR	≤ 2:1	≤ 2:1
Realized Gain	−5 to −3 dBi target	−2 to 0 dBi target
Total Efficiency	25–40% target	50–65% target

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TECHNICAL APPROACH

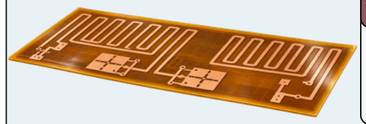
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Single-Feed Spatially Distributed Dual-Band Conformal Antenna for Helmet-Integrated Tactical Communication

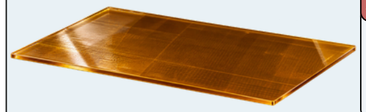
Helmet RF Surface – 6-Layer Integration



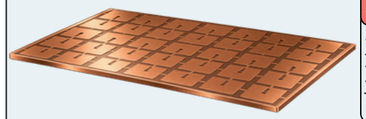
1. Kevlar Helmet Shell
Material: Kevlar | **Thickness:** Existing
Purpose: Ballistic protection + RF environment
Why: **Non-invasive** retrofit and **RF Transparent**



2. Copper RF Radiator
Material: Cu | **Thickness:** 35 µm
Purpose: Front and Rear **UHF + L-band** radiation
Why: Flexible RF **patterning**



3. Kapton dielectric
Material: Kapton | **Thickness:** 50 µm
Purpose: RF dielectric + **flexibility**
Why: Conformal bending



4. Frequency-Selective RF Backing
Material: Cu reflector + AMC | **AMC:** 4 mm / 30 mm cell
Purpose: Pattern control + **back-radiation** reduction
Why: Frequency-selective **RF control**



5. TPU Protective Layer
Material: TPU | **Thickness:** 0.20 mm
Purpose: Insulation + sealing + **abrasion** protection
Why: **0.15–0.25 mm** optimized sweep



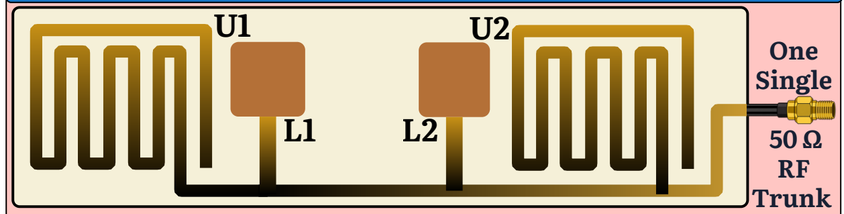
6. Existing Foam Liner
Material: Existing foam | **Thickness:** Existing
Purpose: Comfort + head spacing
Why: No **ergonomic** change

- Low-Profile: 3–5 mm — below Cibin (15 mm) & Michishita (40 mm).
- AMC Trade-off: 9.4–18.7 mm λ/10–λ/20 at 1.6 GHz — 3–5 mm requires CST/HFSS validation.
- Design Target: ≤5 mm, ~23–25 g EST. — literature-benchmarked; prototype verification pending.



TARGET: S11 ≤ −10 dB | VSWR ≤ 2:1 | UHF η 25–40% | L-band η 50–65%

Ultra-Thin Conformal Antenna Layout

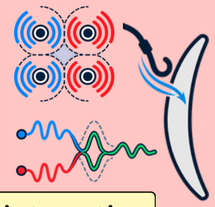


Length: 280 mm | Thickness: ~0.2 mm | Width: 40 mm

Single-Port Passive RF Spine

50 Ω → 1 Coax → Diplexer → Passive **RF Distribution** → **Single RF** Interface

UHF: 50/50 → Balanced | **Front Rear :** 0.35λ
L-band: 80/20 → Tapered | **Front Rear :** 1.28λ
Coherent-Interference **Optimization**



Single coax → Compact, **snag-free** CQB integration

Dual-Band Spatially Distributed Antenna



280 × 40 mm - Flexible, helmet-integrated RF sheet.

403–470 MHz - UHF; λ/4 ≈ 172 mm with a meandered + top-loaded + parasitic radiator.

1.55–1.65 GHz - Compact PIFA for **L-band** operation.

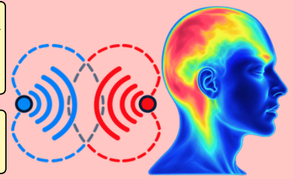
240 mm → **Spatial diversity** • Wider azimuth coverage

Impact: Low-profile • Dual-band • Conformal • No protruding whip

RF Exposure & Back-Radiation Control

Front & Rear : 80/20 split → Controlled **sidelobes** - Reduced localized RF exposure and **interception** footprint.

RF Range: 200–600m UHF + **150-300m** L-band, LOS & NLOS (Target).



AMC Layer for Reduced **back radiation** toward the head.

Low back-radiation • Reduced RF signature • Controlled coverage

Demonstration of End-to-End Tactical Communication & Conformal RF Integration

Audio + Video Input



Voice Capture



Video Capture

Edge Media + Link Control



Opus :Audio
H.264: Video Packetization
freq Hoping

Radio Transmit & Recieve

SDR Transmit Engine



Tx / Rx
Framing
Modulation
Upconversion
Encryption And Filtering

SDR Receiver Engine

RF Down conversion
Signal Demodulation
Filtering
Synchronization

Dual-Band Antenna Surface



Provides conformal UHF and L-band coverage through distributed RF.

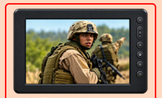
RF Data Confidentiality

Frequency-selective backing targets ≥90% backside RF reduction while preserving coverage.

Audio + Video Output



Voice Output



Video Output

CQB MANET Network



Enables secure, decentralized team communication without fixed infrastructure.

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FEASIBILITY AND VIABILITY

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Technical Feasibility:

- 403–470 MHz UHF — meandered radiator reduces the λ/4 size of ~172mm.
- 280 × 50 mm flexible RF sheet - integrates 4 radiating sections.
- 3–5 mm total profile - compatible with low-profile helmet integration.

Technical Viability:

- 1 × 50 Ω RF port → diplexer → 4 radiators - single-feed architecture.
- 50/50 UHF + 80/20 L-band split - passive power distribution without RF switching.
- S11 ≤ −10 dB / VSWR ≤ 2:1 - defined RF acceptance targets.

Operational Feasibility:

- 240 mm front–rear spacing - supports spatially distributed radiation.
- Flexible copper/Kapton - conforms to helmet curvature.
- No drilling or structural modification - non-invasive integration.

FEASIBILITY & VIABILITY

Operational Viability:

- Front/rear excitation - wider azimuthal coverage.
- AMC + conductive backing - controlled back-radiation.
- Head/helmet/liner loading - accounts for detuning and RF exposure.

Economic Feasibility:

- Passive RF network - no RF switch or active electronics.
- Single RF connection - simpler cabling and integration.
- Flexible printed RF fabrication - supports repeatable manufacturing.

Economic Viability:

- ~0.2 mm RF sheet - lightweight, low-material construction.
- ~23–25 g antenna assembly - minimal added helmet weight.
- Integrated dual-band design - eliminates separate UHF/L-band antennas.

Conventional	Our Solution
External / protruding	Conformal integrated sheet
Single radiator	Front + rear radiators
~172 mm λ/4	Compact meandered UHF
Separate antenna/feed	1 × 50 Ω coax
RF switching	No RF switch
Single-band	Dual-band: 403–470 MHz + 1.55–1.65 GHz
Fixed location	240 mm distribution
Rigid / bulky	Flexible Kapton & copper
Curvature integration difficult	Helmet-conformal
Limited loading analysis	Head/helmet/SAR analysis

VALIDATED TECHNICAL BASIS

1. CRPF, Draft QRs/TDs for UHF Mobile/Base Transceiver, 403–470 MHz.
2. BEL / Department of Defence Production, Secure Digital Handheld Radio-5W, 403–470 MHz, 1/5 W.
3. IEEE MILCOM, “Helmet Mounted UHF Antenna for Mobile User Objective System (MUOS),” 2002.

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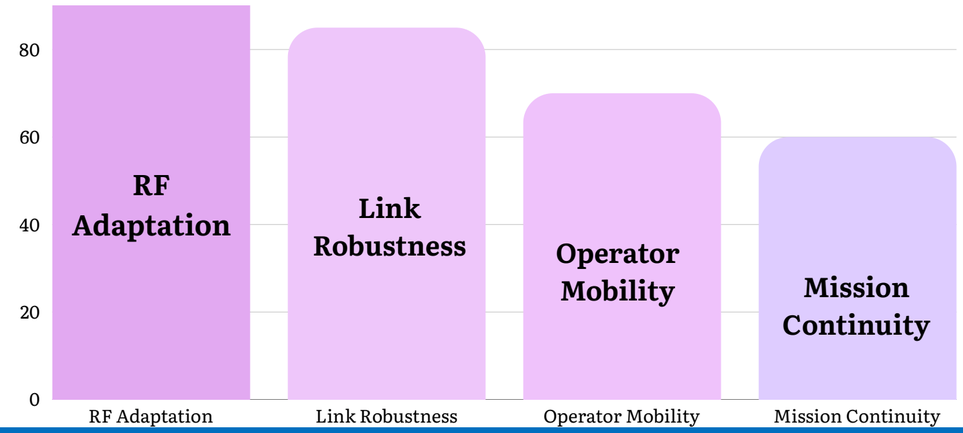
IMPACT AND BENEFITS

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Impact Area	Description	SDG Connection
Enhanced Tactical Communication	Conformal 403–470 MHz UHF antenna integration supports reliable communication in CQB environments.	SDG 3 – Good Health & Well-being. SDG 9 – Industry, Innovation & Infrastructure.
Improved Spatial RF Coverage	Front/rear radiators reduce dependence on a single antenna location under orientation and body-shadowing.	SDG 9 – Industry, Innovation & Infrastructure. SDG 16 – Peace, Justice & Strong Institutions.
Safer Tactical Mobility	Low-profile, helmet-conformal integration minimizes protruding antennas that can interfere with movement or snag on obstacles.	SDG 3 – Good Health & Well-being SDG 11 – Sustainable Cities & Communities
Lightweight Wearable RF Technology	Flexible 280 × 40 mm antenna targets ≤5 mm profile and ≤25 g, supporting compact wearable communication.	SDG 3 – Good Health & Well-being SDG 9 – Industry, Innovation & Infrastructure
Modular Platform Communication	Single 50 Ω feed + passive diplexer/divider network enables a compact dual-band architecture for future RF platforms.	SDG 9 – Industry, Innovation & Infrastructure. SDG 11 – Sustainable Cities & Communities

Metric	Target
Frequency Bands	403–470 MHz + 1.55–1.65 GHz
Antenna Architecture	4 Spatial Radiators
RF Interface	1 Shared 50 Ω Coax
RF Distribution	Passive Diplexer + 50/50 & 80/20 Dividers
Conformal Form Factor	280 × 40 mm, ≤5 mm, ≤25 g

Key benefits of Combat-Aegis



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RESEARCH AND REFERENCES

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Purpose: Reviews conformal helmet antennas for soldier communication.
Source Content: Helmet integration, bandwidth, gain, SAR and metamaterials performance.
Justification: Supports our helmet-integrated 403–470 MHz UHF radiator.

Research Progress on Helmet Antenna in Individual Soldier Communication System

A Dual-Band Wearable Conformal Antenna Based on Artificial Magnetic Conductor

Purpose: Demonstrates flexible dual-band AMC-backed antenna design.
Source Content: Meandered radiator, polyimide substrate, AMC and SAR reduction.
Justification: Supports our flexible L-band AMC-backed PIFA.

Combat Aegis

Purpose: Demonstrates unequal power division using a Wilkinson network.
Source Content: Dual-frequency matching and unequal output power characteristics.
Justification: Supports our 80/20 L-band passive splitter.

A Novel Dual-Frequency Wilkinson Power Divider with Unequal Power Division — 2009.

Front–Back Antenna Diversity for Signal Reliability

Purpose: Studies front/rear antenna placement against body shadowing.
Source Content: Measured impact of human-body obstruction on communication links.
Justification: Supports our front/rear spatial radiator architecture.